

Problem 1 (Operator Norm):

Let $\mathbf{A} \in \mathbb{R}^{m \times n}$ be a rectangular matrix with $m \geq n$.

if you wonder about the difference on supremum and maximum, this post explains it nicely!

- a) Recall the definition of the operator norm of $\mathbf{A}$:

$$\|\mathbf{A}\|_2 = \sup_{\mathbf{x} \in \mathbb{R}^n \setminus \{0\}} \frac{\|\mathbf{A}\mathbf{x}\|}{\|\mathbf{x}\|}.$$

<https://math.stackexchange.com/questions/160451/difference-between-supremum-and-maximum>

Write down the definition of a vector norm. Show that the operator norm is indeed a vector norm in the vector space $\mathbb{R}^{m \times n}$ of matrices.

- b) By the singular value decomposition (SVD) of $\mathbf{A}$, show that $\|\mathbf{A}\|_2 = s_1(\mathbf{A})$, the largest singular value of $\mathbf{A}$.
c) By SVD of $\mathbf{A}$ again, show that $s_1(\mathbf{A}) = \sqrt{\lambda_1(\mathbf{A}^T \mathbf{A})}$, where $\lambda_1(\cdot)$ denotes the largest eigenvalue.

a) Let $\|\cdot\|$ denote an arbitrary norm on some space S , e.g. $\mathbb{R}^n$. Then it is characterized by the following properties:

(i) non-negativity: $\forall x \in \mathbb{R}^n: \|x\| \geq 0$; $\|x\| = 0 \Leftrightarrow x = 0$

(ii) absolute homogeneity: $\forall \alpha \in \mathbb{R}, \forall x \in \mathbb{R}^n: \|\alpha x\| = |\alpha| \cdot \|x\|$

(iii) triangle inequality: $\forall x, y \in \mathbb{R}^n: \|x+y\| \leq \|x\| + \|y\|$

Examples: Vector norms: • Euclidean norm: $\|x\|_2 = \sqrt{x_1^2 + x_2^2 + \dots + x_n^2}$

• Maximum norm: $\|x\|_\infty = \max_i |x_i|$

• L_1 norm: $\|x\|_1 = \sum_{i=1}^n |x_i|$

• L_p norm: $\|x\|_p = \left(\sum_{i=1}^n |x_i|^p \right)^{1/p}$

Let $\mathbf{A} \in \mathbb{R}^{m \times n}$, $m \geq n$. $\frac{\|\mathbf{A}\mathbf{x}\|_2}{\|\mathbf{x}\|_2}$ is a norm.
Show that $\|\mathbf{A}\|_2 := \sup_{\mathbf{x} \in \mathbb{R}^n \setminus \{0\}} \frac{\|\mathbf{A}\mathbf{x}\|_2}{\|\mathbf{x}\|_2}$

(i) non-negativity: $\forall \mathbf{A} \in \mathbb{R}^{m \times n}: \|\mathbf{A}\|_2 \geq 0$, $\|\mathbf{A}\|_2 = 0 \Leftrightarrow \mathbf{A} = 0$

$\sup_{\mathbf{x} \in \mathbb{R}^n \setminus \{0\}} \frac{\|\mathbf{A}\mathbf{x}\|_2}{\|\mathbf{x}\|_2} \geq 0 \checkmark$, $\|\mathbf{A}\|_2 = 0 \Leftrightarrow \sup_{\mathbf{x} \in \mathbb{R}^n \setminus \{0\}} \|\mathbf{A}\mathbf{x}\|_2 = 0 \Leftrightarrow \mathbf{A} = 0 \checkmark$
 > 0 (since $\mathbf{x} \neq 0$)

$\uparrow$
we can choose any $\mathbf{x}$ here and want to maximize $\|\mathbf{A}\mathbf{x}\|$.
If any col of $\mathbf{A}$ contains non-zero entries, we can choose the corresponding row of $\mathbf{x}$ to be non-zero too!

$$(ii) \alpha \in \mathbb{R}, \quad \|\alpha \cdot A\|_2 = \sup_{x \in \mathbb{R}^n \setminus \{0\}} \frac{\|\alpha Ax\|_2}{\|x\|_2} = \sup_{x \in \mathbb{R}^n \setminus \{0\}} \frac{|\alpha| \cdot \|Ax\|_2}{\|x\|_2} = |\alpha| \cdot \sup_{x \in \mathbb{R}^n \setminus \{0\}} \frac{\|Ax\|_2}{\|x\|_2} = |\alpha| \cdot \|A\|_2$$

$$(iii) \|A+B\|_2 = \sup_{x \in \mathbb{R}^n \setminus \{0\}} \frac{\|(A+B)x\|_2}{\|x\|_2} \leq \sup_{x \in \mathbb{R}^n \setminus \{0\}} \frac{\|Ax\|_2 + \|Bx\|_2}{\|x\|_2} \leq \sup_{x \in \mathbb{R}^n \setminus \{0\}} \frac{\|Ax\|_2}{\|x\|_2} + \sup_{x \in \mathbb{R}^n \setminus \{0\}} \frac{\|Bx\|_2}{\|x\|_2} \quad \checkmark$$

sometimes also written as $\sigma_{\max}$

$$= \|A\|_2 + \|B\|_2$$

b) Show: $\|A\|_2 = s_1(A)$

SVD of A: $A = U \Sigma V^T$ w/ $U \in \mathbb{R}^{m \times m}$, $\Sigma \in \mathbb{R}^{m \times n}$, $V \in \mathbb{R}^{n \times n}$

$$\& U^T U = I \\ W^T = I$$

$$\Sigma = \begin{bmatrix} s_1 & & & \\ & s_2 & & \\ & & \ddots & \\ & & & s_n \\ & & & & 0 \end{bmatrix} \cdot \begin{bmatrix} y_1 \\ \vdots \\ y_n \\ 0 \end{bmatrix} = \begin{bmatrix} s_1 y_1 \\ s_2 y_2 \\ \vdots \\ s_n y_n \\ 0 \end{bmatrix}$$

Consider two vectors a & b ,
then $\frac{\| \Sigma a \|_2}{\|a\|_2} = \frac{\| \Sigma \cdot s_1 e_1 \|_2}{\|s_1 e_1\|_2} = \frac{\| \Sigma e_1 \|_2}{\|e_1\|_2}$

Proof:

$$\sup_{x \in \mathbb{R}^n \setminus \{0\}} \frac{\|Ax\|_2}{\|x\|_2} = \sup_{x \in \mathbb{R}^n \setminus \{0\}} \frac{\|U \Sigma V^T x\|_2}{\|x\|_2} = \sup_{x \in \mathbb{R}^n \setminus \{0\}} \frac{\|U \Sigma \underbrace{V^T x}_y\|_2}{\|x\|_2} = \sup_{y \in \mathbb{R}^n \setminus \{0\}} \frac{\|U \Sigma y\|_2}{\|y\|_2} = \sup_{y \in \mathbb{R}^n \setminus \{0\}} \frac{\|\Sigma y\|_2}{\|y\|_2} = \sup_{\|y\|_2=1} \|\Sigma y\|_2$$

$$\|V^T x\|_2 = \|x\|_2$$

Since $\|V^T x\|_2^2 = (V^T x)^T (V^T x) = x^T V^T V x = x^T x = \|x\|_2^2$

$$= \sup_{\|y\|_2=1} \left\| \begin{pmatrix} s_1 y_1 \\ s_2 y_2 \\ \vdots \\ s_n y_n \end{pmatrix} \right\|_2$$

$$= s_1$$

Supremum achieved with

$$y = \begin{pmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{pmatrix}$$

c) Show: $s_1(A) = \sqrt{\lambda_1(A^T A)}$

Proof:

SVD: $A = U \Sigma V^T \Rightarrow A^T A = (U \Sigma V^T)^T (U \Sigma V^T)$

$$= V \Sigma^T U^T U \Sigma V^T$$

$$= V \Sigma^T \Sigma V^T$$

$A^T A$ is symmetric
and thus diagonalizable

Eigenvalue decomposition of $A^T A = V D V^T$

$$\left\{ \begin{aligned} D &= \Sigma^T \Sigma \in \mathbb{R}^{n \times n} \\ &= \begin{bmatrix} s_1^2 & & & \\ & s_2^2 & & \\ & & \ddots & \\ & & & s_n^2 \\ & & & & 0 \end{bmatrix} \cdot \begin{bmatrix} s_1 & s_2 & & \\ & 0 & \ddots & s_n \\ & & & 0 \end{bmatrix} = \begin{bmatrix} s_1^2 & & & \\ & s_2^2 & & \\ & & \ddots & \\ & & & s_n^2 \end{bmatrix} \end{aligned} \right.$$

$$s_1^2 \geq s_2^2 \geq \dots \geq s_n^2$$

$$\downarrow$$

$$\lambda_1(A^T A) = s_1^2$$

$$\Leftrightarrow s_1 = \sqrt{\lambda_1(A^T A)}$$

Problem 2 (Calculus):

- a) Let $\mathbf{A} \in \mathbb{R}^{m \times n}$ be a rectangular matrix, $\mathbf{x} \in \mathbb{R}^n, \mathbf{y} \in \mathbb{R}^m$ be column vectors. Let $f(\mathbf{x}) = \|\mathbf{Ax} - \mathbf{y}\|_2^2$. Compute its gradient $\nabla f \in \mathbb{R}^n$.
- b) Let $\mathbf{U} \in \mathbb{R}^{n \times p}, \mathbf{V} \in \mathbb{R}^{p \times m}$, and $\mathbf{R} \in \mathbb{R}^{n \times m}$. The matrix factorization problem tries to find an approximation of $\mathbf{R}$ as the product of two matrices with smaller common dimension p , that is

$$\mathbf{R} \approx \mathbf{UV}.$$

This problem can be solved for instance by minimizing the loss $L(\mathbf{U}, \mathbf{V}) := \frac{1}{2} \|\mathbf{UV} - \mathbf{R}\|_F^2$, where $\|\mathbf{A}\|_F := \sqrt{\sum_i \sum_j |A_{ij}|^2}$ is the Frobenius norm.

Compute the derivative of L w.r.t. $\mathbf{U}$, $\frac{\partial L}{\partial \mathbf{U}}$, and w.r.t. $\mathbf{V}$, $\frac{\partial L}{\partial \mathbf{V}}$, respectively.

- c) Consider the problem of non-linear least squares regression with some non-linear function $\ell: \mathbb{R} \rightarrow \mathbb{R}$ and n data samples $(\mathbf{x}_i, y_i)$,

$$L(\mathbf{w}) = \frac{1}{2n} \sum_{i=1}^n (y_i - \ell(\mathbf{x}_i^\top \mathbf{w}))^2.$$

Compute the gradient $\nabla_{\mathbf{w}} L(\mathbf{w})$ and the Hessian $\nabla_{\mathbf{w}}^2 L(\mathbf{w})$.

1) $f(\mathbf{x}) = \|\mathbf{Ax} - \mathbf{y}\|_2^2, \quad \mathbf{x} \in \mathbb{R}^n$

1) we know $\nabla f(\mathbf{x}) \in \mathbb{R}^n$. This is always a good sanity check!

2) Two ways to compute:

• directly applying chain rule + vector calculus:

$$\nabla f(\mathbf{x}) = 2 \underbrace{\mathbf{A}^\top (\mathbf{Ax} - \mathbf{y})}_{n \times n \quad m \times n \quad n \quad m}$$

• compute $\partial_{x_p} f(\mathbf{x})$.

$$\begin{aligned} \text{First rewrite } f(\mathbf{x}) &= \|\mathbf{Ax} - \mathbf{y}\|_2^2 = \sum_{i=1}^m \left((\mathbf{Ax})_i - y_i \right)^2 \\ &= \sum_{i=1}^m \left(\sum_{j=1}^n A_{ij} \cdot x_j - y_i \right)^2 \end{aligned}$$

$$\partial_{x_p} f(\mathbf{x}) = \partial_{x_p} \left[\sum_{i=1}^m \left(\sum_{j=1}^n A_{ij} \cdot x_j - y_i \right)^2 \right]$$

$$= \sum_{i=1}^m 2A_{ip} \cdot \left(\sum_{j=1}^n A_{ij} x_j - y_i \right)$$

$$= 2 \sum_{i=1}^m A_{ip} \cdot [(Ax)_i - y_i]$$

Let a_p denote columns of $A = \begin{bmatrix} | & & | \\ a_1 & \dots & a_n \\ | & & | \end{bmatrix}$

$$= 2 a_p^T \cdot (Ax - y)$$

→ sanity check: $y \in \mathbb{R}^m$, $a_p \in \mathbb{R}^m$ ✓
same dimension

$$\Rightarrow \begin{bmatrix} \partial_{x_1} f(x) \\ \vdots \\ \partial_{x_n} f(x) \end{bmatrix} = \nabla f(x) = 2 \begin{bmatrix} a_1^T \\ \vdots \\ a_n^T \end{bmatrix} \cdot (Ax - y)$$

$$= 2A^T (Ax - y)$$

$$b) L(U, V) = \frac{1}{2} \|UV - R\|_F^2 \quad U \in \mathbb{R}^{n \times p}, V \in \mathbb{R}^{p \times m}$$

Two ways to compute it:

1) matrix calculus:

$$(i) \text{ we know: } \frac{\partial}{\partial X} \|X\|_F^2 = \frac{\partial}{\partial X} \text{Tr}(XX^H) = 2X$$

$$(ii) \text{ rewrite } L(U, V) \text{ as: } L(U, V) = \frac{1}{2} \|Z(U, V)\|_F^2$$

$$\text{Then } \frac{\partial L(U, V)}{\partial U} = \frac{\partial L(U, V)}{\partial Z} \cdot \frac{\partial Z}{\partial U}$$

$$\frac{\partial L(U, V)}{\partial V} = \frac{\partial L(U, V)}{\partial Z} \cdot \frac{\partial Z}{\partial V}$$

! what makes it tricky is that $Z \in \mathbb{R}^{n \times m}$, so $\frac{\partial Z}{\partial U} \in \mathbb{R}^{(n \times m) \times (n \times p)}$

$\frac{\partial L}{\partial z} \in \mathbb{R}^{(n \times m)}$, so how do we "multiply"?

⇒ actually, what one should do is to apply the chain rule entry-wise:

$$\frac{\partial L(u, v)}{\partial v_{ij}} = \sum_{k, l} \frac{\partial L(u, v)}{\partial z_{k, l}} \cdot \frac{\partial z_{k, l}}{\partial v_{ij}}$$

→ Tensor calculus: you can read more here:

<https://cs231n.stanford.edu/vecDerivs.pdf>

Alternatively: compute derivatives entry-wise again:

$$L(u, v) = \frac{1}{2} \|uv - R\|_F^2 = \frac{1}{2} \sum_{i=1}^n \sum_{j=1}^m (uv_{ij} - R_{ij})^2 = \frac{1}{2} \sum_{i=1}^n \sum_{j=1}^m \left(\sum_{k=1}^p u_{ik} v_{kj} - R_{ij} \right)^2$$

$$\sim \frac{\partial}{\partial u_{qr}} L(u, v) = \sum_{j=1}^m (uv_{qj} - R_{qj}) \cdot v_{rj} = ((uv - R) \cdot v^T)_{qr}$$

$$\sim \frac{\partial}{\partial v_{qr}} L(u, v) = \sum_{i=1}^n (uv_{ir} - R_{ir}) \cdot u_{iq} = (u^T (uv - R))_{qr}$$

reshaping leads to: $\nabla_u L(u, v) = \underbrace{(uv - R)}_{n \times m} \cdot \underbrace{v^T}_{m \times p}$ sanity check ✓

$$\nabla_v L(u, v) = \underbrace{u^T}_{p \times n} \underbrace{(uv - R)}_{n \times m}$$
 sanity check ✓

! interestingly this is exactly what we get if we would extend the rules from vector calculus to matrix calculus. But one needs to be careful !

$$c) L(w) = \frac{1}{2n} \sum_{i=1}^n (y_i - \ell(w^T x_i))^2$$

chain rule: $L(w) = \frac{1}{2n} \sum_{i=1}^n g(\ell(z(w)))$

$$\nabla_w L(w) = -\frac{1}{n} \sum_{i=1}^n (y_i - \ell(w^T x_i)) \cdot \ell'(w^T x_i) \cdot x_i$$

with $z(w) = w^T x_i$
 $g(a) = (y_i - a)^2$

$$\nabla_w^2 L(w) = \frac{1}{n} \sum_{i=1}^n (\ell'(w^T x_i))^2 \cdot x_i x_i^T + (y_i - \ell(w^T x_i)) \cdot \ell''(w^T x_i) x_i x_i^T$$

product rule & chain rule

$$= \frac{1}{n} \sum_{i=1}^n \left[(\ell'(w^T x_i))^2 + (y_i - \ell(w^T x_i)) \cdot \ell''(w^T x_i) \right] x_i x_i^T$$

Problem 3 (Taylor Expansion):

Let $f: \mathbb{R}^n \rightarrow \mathbb{R}$ be a twice differentiable function with a local minimum $\mathbf{x}^*$.

- a) Write down the definition of the Hessian $\mathbf{H}(\mathbf{x}) = \nabla^2 f(\mathbf{x}) \in \mathbb{R}^{n \times n}$ of f and the order-2 Taylor expansion of f at $\mathbf{x}^*$.
- b) Using Chain rule or otherwise, prove that for any $\mathbf{x}, \mathbf{v} \in \mathbb{R}^n$, the matrix-vector product $\mathbf{H}(\mathbf{x})\mathbf{v} \in \mathbb{R}^n$ is equal to a limit:

$$\mathbf{H}(\mathbf{x})\mathbf{v} = \lim_{t \rightarrow 0} \frac{\nabla f(\mathbf{x} + t\mathbf{v}) - \nabla f(\mathbf{x})}{t}.$$

- c) Suppose f is a L -smooth function, that is, there is an $L > 0$ such that

$$\|\nabla f(\mathbf{x}_1) - \nabla f(\mathbf{x}_2)\| \leq L \|\mathbf{x}_1 - \mathbf{x}_2\|, \forall \mathbf{x}_1, \mathbf{x}_2 \in \mathbb{R}^n.$$

Show that each eigenvalue of $\mathbf{H}(\mathbf{x})$ of any $\mathbf{x} \in \mathbb{R}^n$ is upper bounded by L .

- d) Recall that $\mathbf{x}^*$ is a local minimum. Using Problem 1c) and 3c) or otherwise, prove that

$$\|\mathbf{H}(\mathbf{x}^*)\|_2 \leq L$$

- e) Using Cauchy-Schwartz inequality and Problem 3d) or otherwise, show that

$$f(\mathbf{x}) - f(\mathbf{x}^*) \leq \frac{L}{2} \|\mathbf{x} - \mathbf{x}^*\|^2 + o(\|\mathbf{x} - \mathbf{x}^*\|^3).$$

$$a) \mathbf{H}(\mathbf{x}) = \nabla^2 f(\mathbf{x}) \in \mathbb{R}^{n \times n}.$$

For every entry: $(\mathbf{H}(\mathbf{x}))_{ij} = \frac{\partial^2}{\partial x_i \partial x_j} f(\mathbf{x})$ with $\mathbf{x} = \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix}$

sometimes written as:

Order-2 Taylor expansion:

$$\langle \nabla f(\mathbf{x}^*), \mathbf{x}^* - \mathbf{x} \rangle$$

$$f(\mathbf{x}) = f(\mathbf{x}^*) + \nabla f(\mathbf{x}^*)^T (\mathbf{x} - \mathbf{x}^*) + \frac{1}{2} (\mathbf{x} - \mathbf{x}^*)^T \mathbf{H}(\mathbf{x}^*) (\mathbf{x} - \mathbf{x}^*) + o(\|\mathbf{x} - \mathbf{x}^*\|^3)$$

$\nabla f(\mathbf{x}^*) = 0$

$$= f(\mathbf{x}^*) + \frac{1}{2} (\mathbf{x} - \mathbf{x}^*)^T \mathbf{H}(\mathbf{x}^*) (\mathbf{x} - \mathbf{x}^*) + o(\|\mathbf{x} - \mathbf{x}^*\|^3)$$

b) Show: $H(x)v = \lim_{t \rightarrow 0} \frac{\nabla f(x+tv) - \nabla f(x)}{t}$

Note that:

$$\lim_{t \rightarrow 0} \frac{\nabla f(x+tv) - \nabla f(x)}{t} = \frac{\partial}{\partial x} \underbrace{(\nabla f(x)^T v)}_{\sum_{i=1}^n \partial_{x_i} f(x) v_i} = \underbrace{\left[\frac{\partial}{\partial x} \nabla f(x) \right]}_{H(x)} v + \nabla f(x)^T \underbrace{\left[\frac{\partial}{\partial x} v \right]}_{=0} = H(x)v$$

where does the transpose go? ∇

$$\left[\lim_{t \rightarrow 0} \frac{\partial_{x_1} f(x+tv) - \partial_{x_1} f(x)}{t} \right] \begin{bmatrix} \frac{\partial}{\partial x_1} \\ \vdots \\ \frac{\partial}{\partial x_n} \end{bmatrix}$$

Let $h(t) = x + tv$, and $g: \mathbb{R} \rightarrow \mathbb{R}$, then $\lim_{t \rightarrow 0} \frac{g(x+tv) - g(x)}{t} = \frac{\partial}{\partial t} g(h(t)) \Big|_{t=0} = \nabla g(h(t)) \cdot h'(t) \Big|_{t=0} = \nabla g(x)^T v$

with $g(x) = \partial_{x_i} f(x)$, $\nabla g(x)^T$ corresponds to the i th row of $H(x)$

c) for each eigenvalue λ_i & corresponding eigenvector v_i it holds that

$$H(x) \cdot v_i = \lambda_i \cdot v_i$$

For L -smooth functions: $\|\nabla f(x) - \nabla f(y)\| \leq L \cdot \|x - y\| \quad \forall x, y \in \mathbb{R}^n$

Proof: For any eigenvalue λ_i of $H(x)$:

$$\lambda_i = \|\lambda_i v_i\| = \|H(x) \cdot v_i\| = \lim_{t \rightarrow 0} \left\| \frac{\nabla f(x + v_i t) - \nabla f(x)}{t} \right\|$$

$$\stackrel{\substack{\uparrow \\ 3b)}}{=} \lim_{t \rightarrow 0} \frac{\|\nabla f(x + v_i t) - \nabla f(x)\|}{t} \leq \lim_{t \rightarrow 0} \frac{L \cdot \|v_i t\|}{t} = L \cdot \|v_i\| = L$$

continuity of $\|\cdot\|$ in $\mathbb{R}^n \rightarrow \mathbb{R}$

□

d) Show: $\|H(x^*)\|_2 \leq L$

from 1c): $\|H(x^*)\|_2 = s_1(H(x^*)) = \sqrt{\lambda_1(H(x^*)^T H(x^*))}$
why?
 $\downarrow$
 $= \lambda_1(H(x^*))$

from 3c): $\lambda_1(H(x^*)) \leq L$ for L -smooth func.

$$\Rightarrow \|H(x^*)\|_2 \leq L$$

e) From 3a): $f(x) = f(x^*) + \underbrace{\frac{1}{2}(x - x^*)^T H(x^*) \cdot (x - x^*)}_{\leq \|H(x^*)\|_2 \cdot \|x^* - x\|_2^2} + o(\|x - x^*\|^3)$
 $\leq L \cdot \|x^* - x\|_2^2$
 $\Rightarrow |f(x) - f(x^*)| \leq \frac{L}{2} \|x - x^*\|_2^2 + o(\|x - x^*\|^3)$

Problem 4 (Expectations, conditioning, and variance):

Let X, Y be discrete random variables with finite second moments.

a) Show that

$$\text{Var}(X) = \mathbb{E}[X^2] - \mathbb{E}[X]^2.$$

b) Starting from

$$P(X = x) = \sum_y P(X = x \mid Y = y)P(Y = y),$$

prove $\mathbb{E}[X] = \mathbb{E}_Y[\mathbb{E}[X \mid Y]]$.

c) Prove

$$\text{Var}_X(X) = \mathbb{E}_Y[\text{Var}(X \mid Y)] + \text{Var}_Y(\mathbb{E}[X \mid Y]).$$

Here, $\mathbb{E}[X \mid Y]$ and $\text{Var}(X \mid Y)$ should be viewed as functions of the random variable Y .

d) Finally, let $Y \sim \text{Bernoulli}(p)$, and suppose

$$\mathbb{E}[X \mid Y = i] = \mu_i, \quad \text{Var}(X \mid Y = i) = \sigma_i^2, \quad i \in \{0, 1\}.$$

Compute $\mathbb{E}[X]$ and $\text{Var}(X)$. Identify which part of $\text{Var}(X)$ describes variability within the two conditional distributions and which part describes variability between their conditional means. What happens to these two terms if (i) $\mu_0 = \mu_1$, or (ii) $\sigma_0^2 = \sigma_1^2 = 0$?

a) Using the definition of variance & simplifying we get:

$$\begin{aligned} \text{Var}(X) &= \mathbb{E}[(X - \mathbb{E}[X])^2] = \mathbb{E}[X^2 - 2X\mathbb{E}[X] + \mathbb{E}[X]^2] \\ &= \mathbb{E}[X^2] - 2\mathbb{E}[X]^2 + \mathbb{E}[X]^2 \\ &= \mathbb{E}[X^2] - \mathbb{E}[X]^2 \quad \square \end{aligned}$$

b) We use the definition of $\mathbb{E}[X]$ for discrete r.v. & the law of total probability:

$$\mathbb{E}[X] = \sum_x x P(X=x) = \sum_x x \sum_y P(X=x \mid Y=y) P(Y=y)$$

rearranging the sum gives

$$\begin{aligned} &= \sum_y P(Y=y) \underbrace{\sum_x x P(X=x \mid Y=y)}_{\mathbb{E}[X \mid Y=y]} \\ &= \sum_y P(Y=y) \mathbb{E}[X \mid Y=y] \\ &= \mathbb{E}_Y[\mathbb{E}[X \mid Y]] \end{aligned}$$

Law of total variance.

c)

c) Prove

$$\text{Var}_X(X) = \mathbb{E}_Y[\text{Var}(X | Y)] + \text{Var}_Y(\mathbb{E}[X | Y]).$$

Here, $\mathbb{E}[X | Y]$ and $\text{Var}(X | Y)$ should be viewed as functions of the random variable Y .

Using results from before we get:

$$\text{Var}_X(X) \stackrel{a)}{=} \mathbb{E}[X^2] - \mathbb{E}[X]^2$$

$$\stackrel{b)}{=} \mathbb{E}_Y[\mathbb{E}[X^2 | Y]] - (\mathbb{E}_Y[\mathbb{E}[X | Y]])^2$$

$$= (\mathbb{E}_Y[\mathbb{E}[X^2 | Y]] - \mathbb{E}_Y[\mathbb{E}[X | Y]^2]) + (\mathbb{E}_Y[\mathbb{E}[X | Y]^2] - (\mathbb{E}_Y[\mathbb{E}[X | Y]])^2)$$

$$= \mathbb{E}_Y[\text{Var}(X | Y)] + \text{Var}_Y(\mathbb{E}[X | Y])$$

d) Finally, let $Y \sim \text{Bernoulli}(p)$, and suppose

$$\mathbb{E}[X | Y = i] = \mu_i, \quad \text{Var}(X | Y = i) = \sigma_i^2, \quad i \in \{0, 1\}.$$

Compute $\mathbb{E}[X]$ and $\text{Var}(X)$. Identify which part of $\text{Var}(X)$ describes variability within the two conditional distributions and which part describes variability between their conditional means. What happens to these two terms if (i) $\mu_0 = \mu_1$, or (ii) $\sigma_0^2 = \sigma_1^2 = 0$?

$$d) \mathbb{E}[X] \stackrel{b)}{=} \mathbb{E}_Y[\mathbb{E}[X | Y]]$$

$$= p \mathbb{E}[X | Y=1] + (1-p) \mathbb{E}[X | Y=0]$$

$$= p \cdot \mu_1 + (1-p) \mu_0$$

$$\text{Var}(X) \stackrel{c)}{=} \underbrace{\mathbb{E}_Y[\text{Var}(X | Y)]}_{\text{variability within condit. distr.}} + \underbrace{\text{Var}_Y(\mathbb{E}[X | Y])}_{\text{variability between condit. means}}$$

$$= p \text{Var}(X | Y=1) + (1-p) \text{Var}(X | Y=0)$$

$$= p \cdot \sigma_1^2 + (1-p) \sigma_0^2$$

$$= \mathbb{E}_Y[\mathbb{E}[X | Y]^2] - \underbrace{\mathbb{E}_Y[\mathbb{E}[X | Y]]^2}_{\mathbb{E}[X]^2}$$

$$= p \mu_1^2 + (1-p) \mu_0^2 - (p \mu_1 + (1-p) \mu_0)^2$$

$$= \dots = p(1-p)(\mu_1 - \mu_0)^2$$

$\Rightarrow$ when $\mu_0 = \mu_1$: $\text{Var}_Y(\mathbb{E}[X|Y]) = 0 \leadsto \text{Var}(X)$ fully determined by variability within conditional distributions.

when $\sigma_0^2 = \sigma_1^2 = 0$: $\mathbb{E}_Y[\text{Var}(X|Y)] = 0 \leadsto \text{Var}(X)$ fully determined by var. between conditional means.